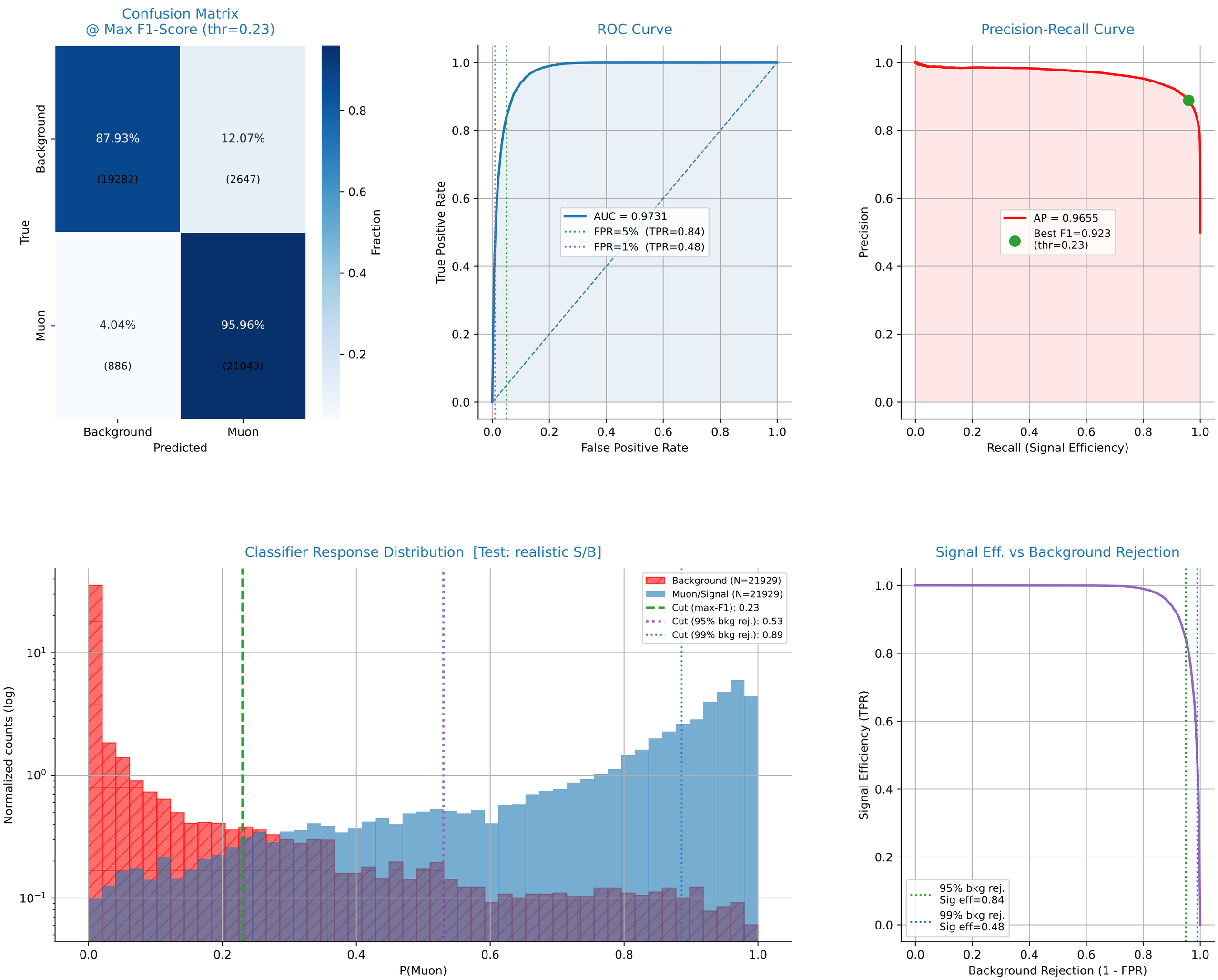
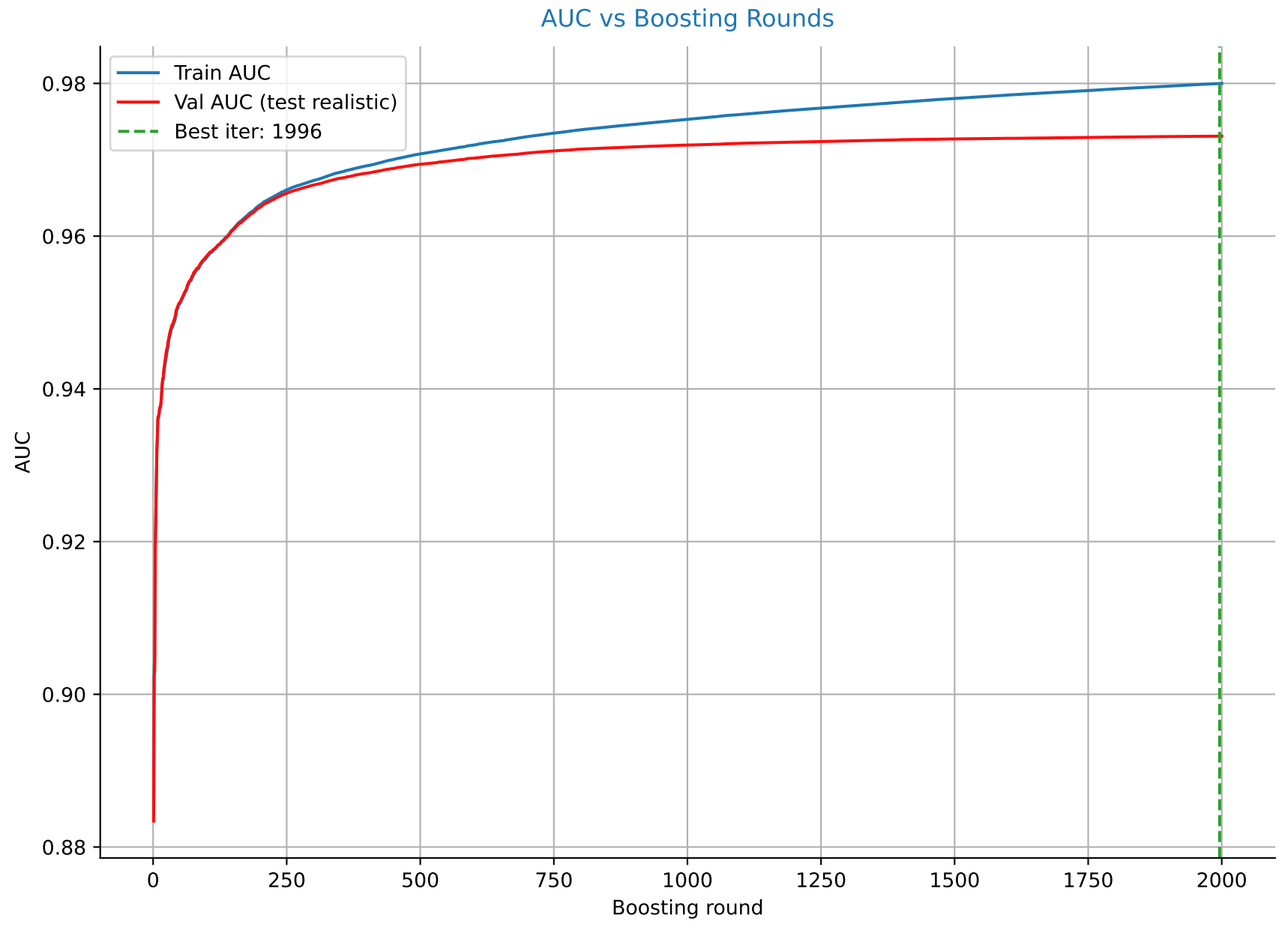
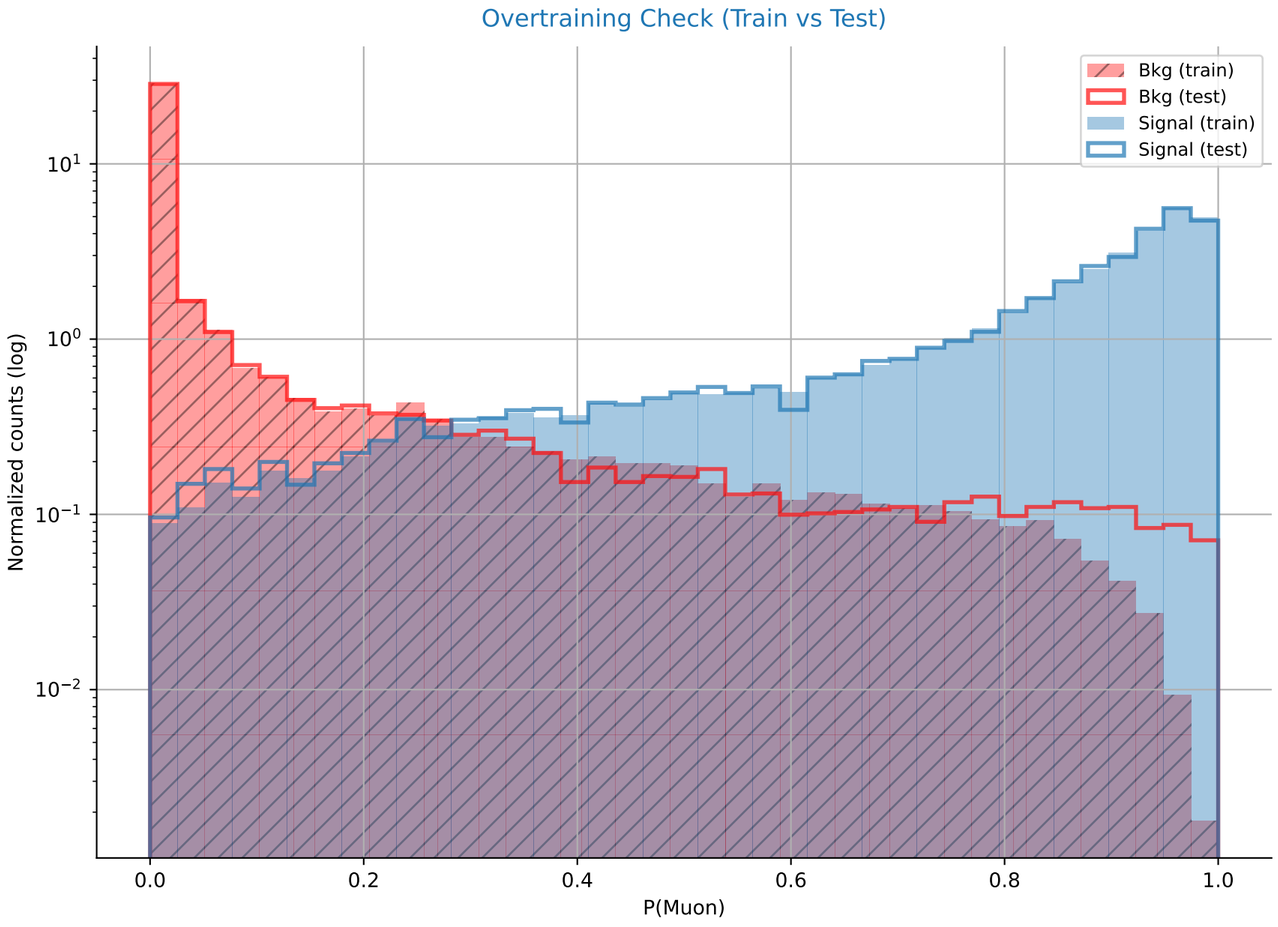
Muon Identification — XGBoost Analysis
[Train S/B=1:0 | Test S/B=1:1 (realistic)]

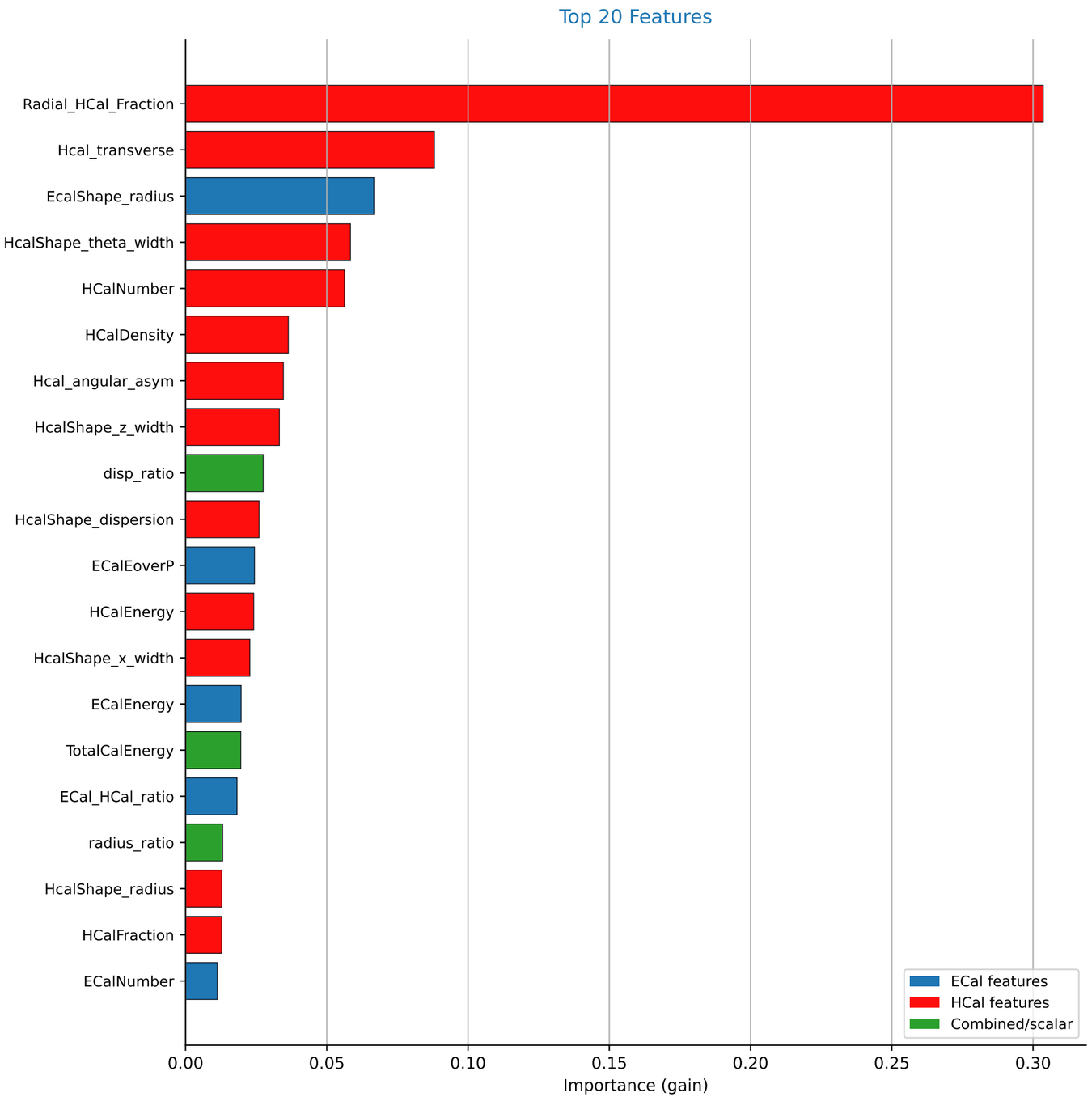


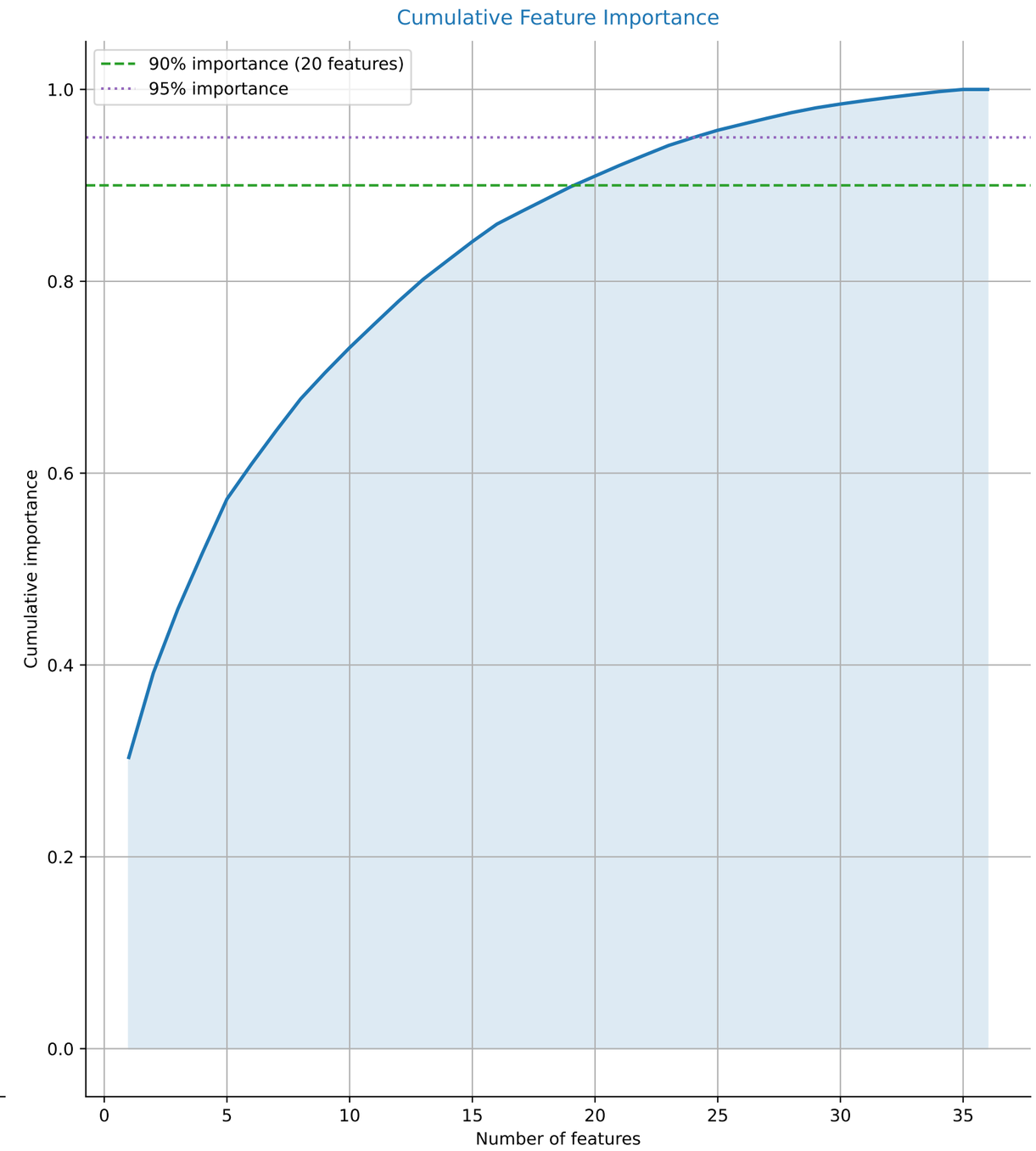
XGBoost Training Diagnostics



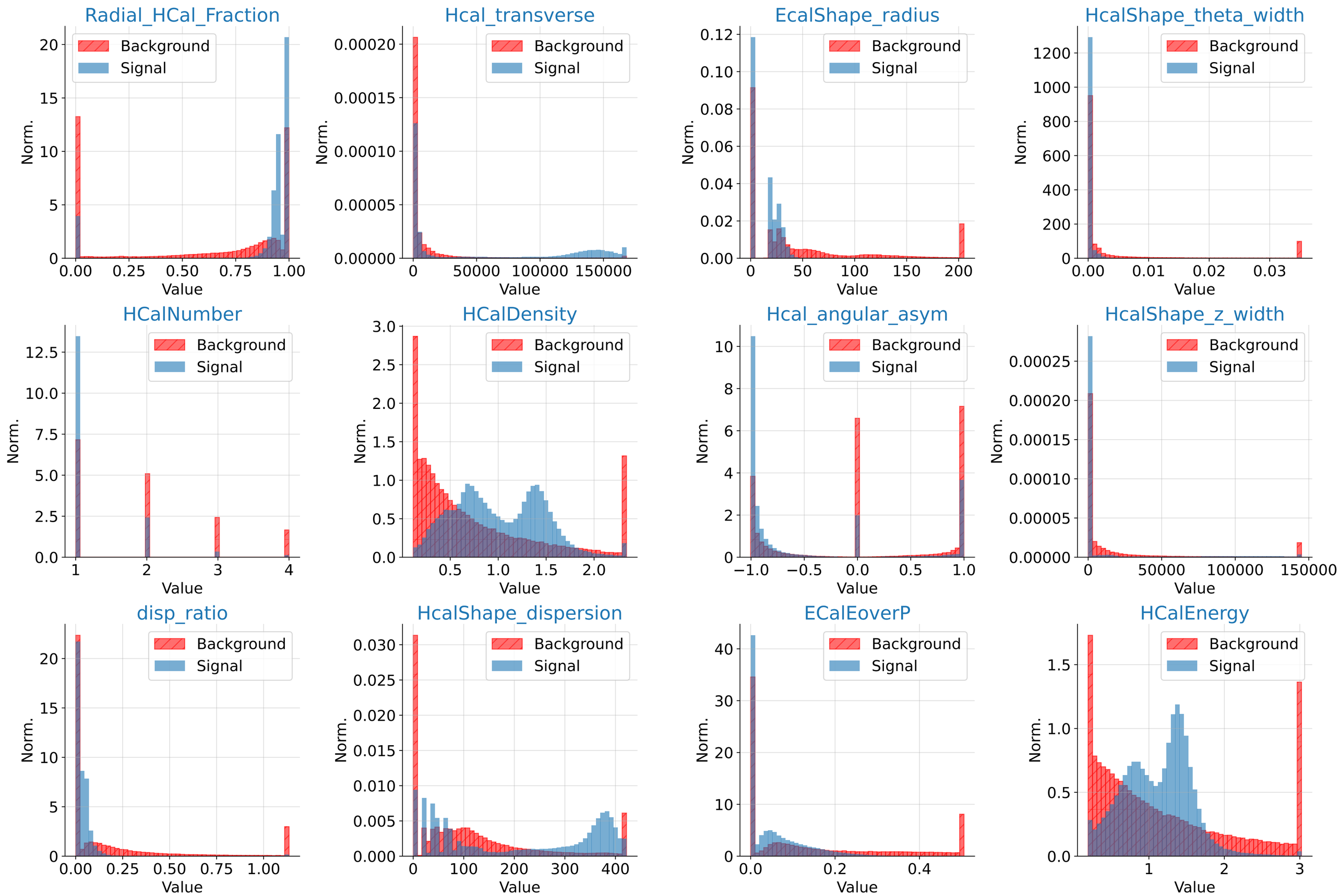


Feature Importances — XGBoost



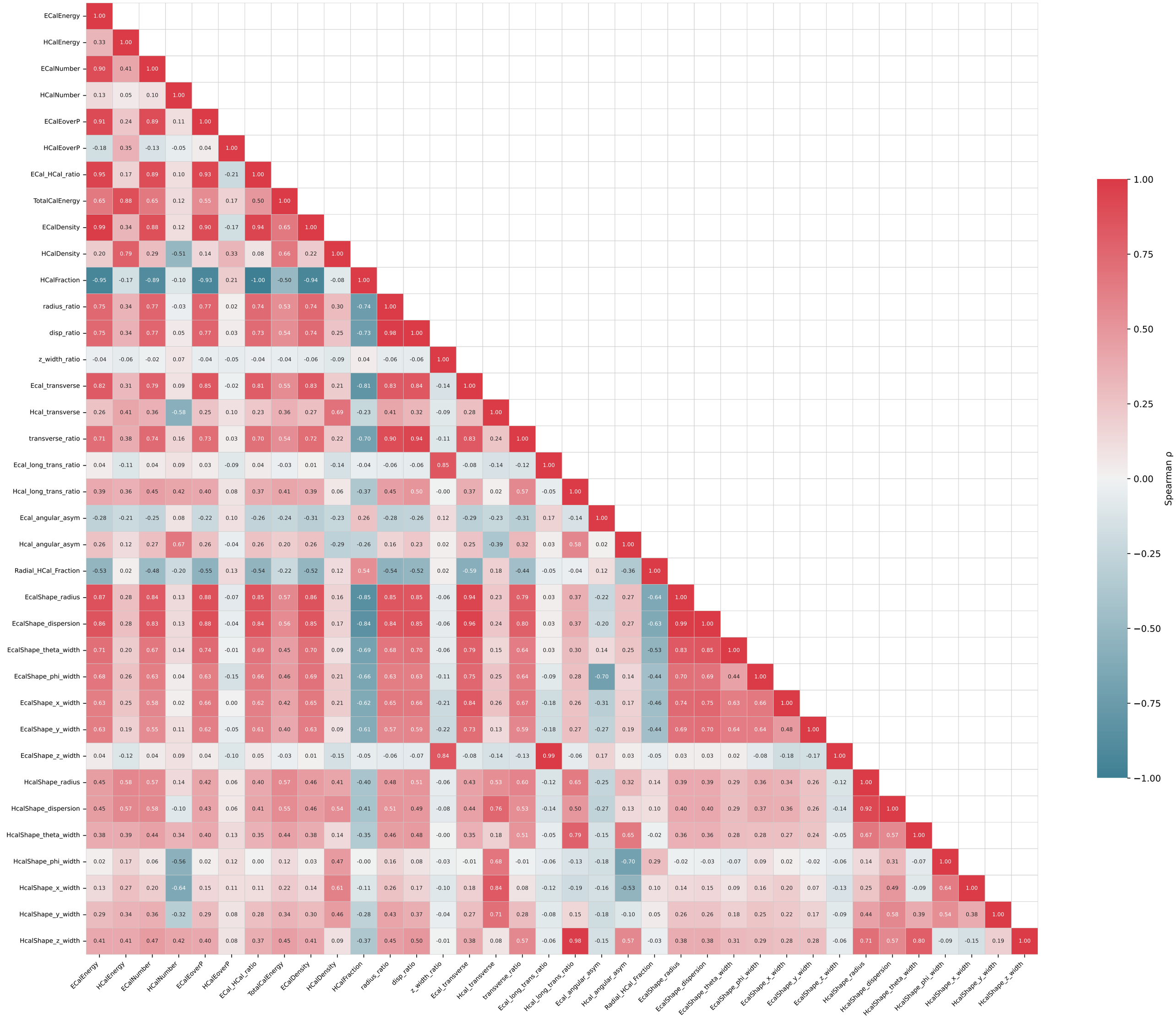


Key Physics Variable Distributions: Signal vs Background
[rozkłady z całego datasetu przed podziałem]

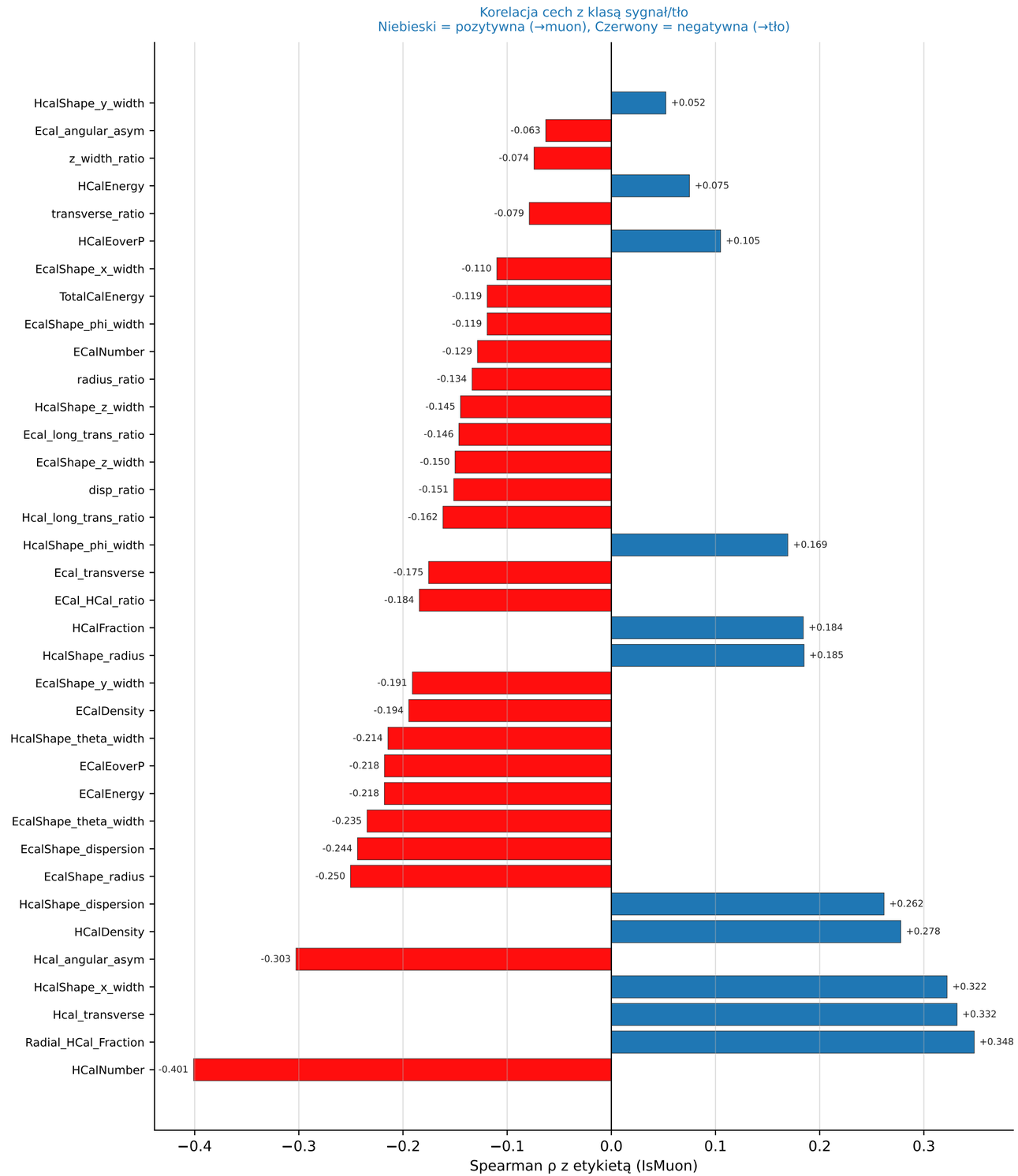


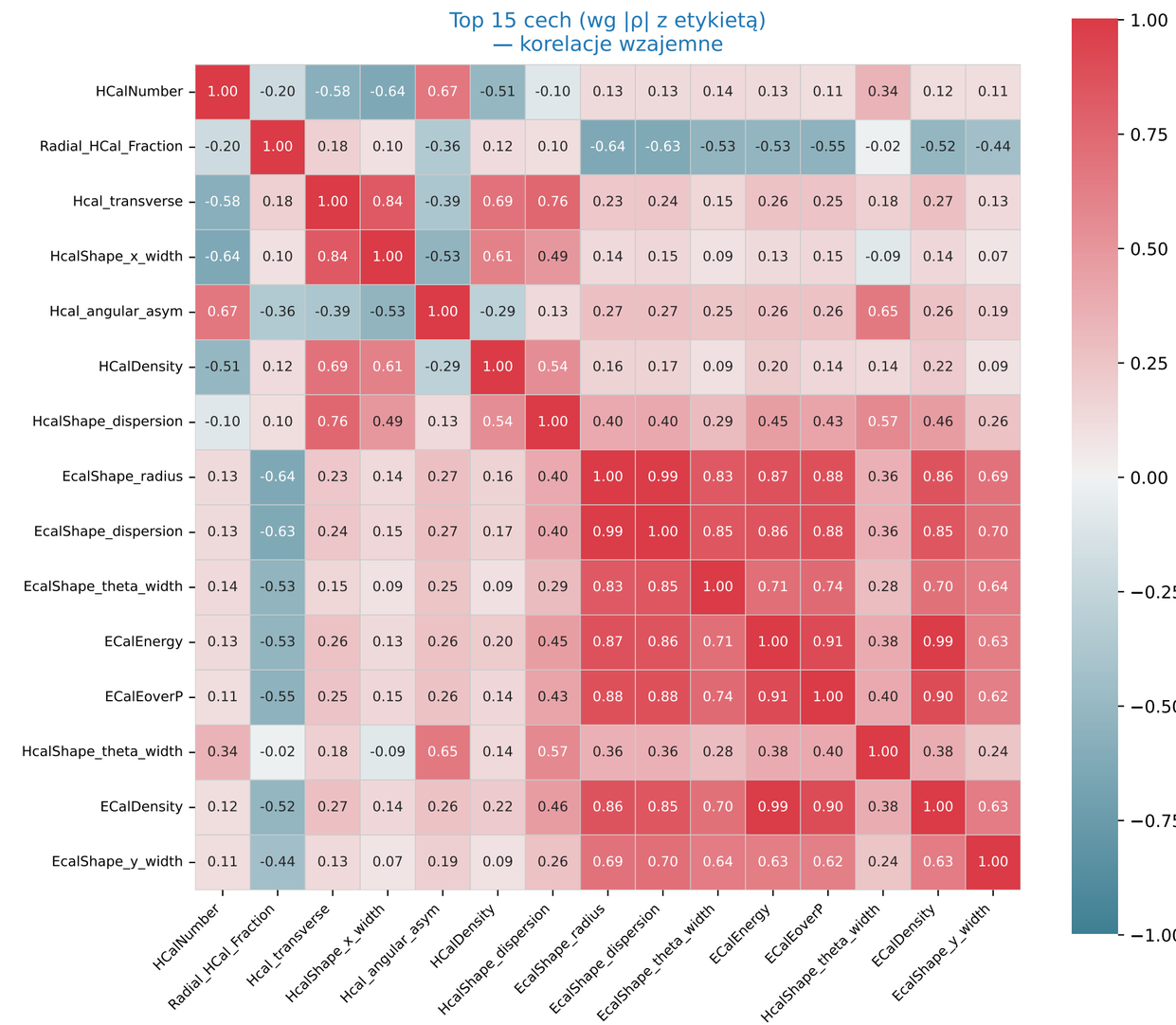
Spearman Correlation Matrix — All Features
(wartości z całego datasetu)

Wyższe |ρ| → silniejsza monotonna zależność między cechami



Feature–Label Correlations & Clustered Heatmap





```
╔═══════════════════════════════════════════════════════════════════════╗
║           Physics Analysis — Muon Identification Report             ║
║                          [ XGBoost ]                                ║
╚═══════════════════════════════════════════════════════════════════════╝

  Input file      : MLDataGrape.root
  Total events    : 431753
  Signal (muons)  : 322105  (74.6%)
  Background      : 109648  (25.4%)
  Features used   : 36

────────────────────────────────────────────────────────────────────────

  STRATEGIA PODZIAŁU:
    Trening  : 80% sygnału + 80% tła   →  S/B = 1:0.3
               Sygnał=257684  Tło=87719
    Test     : 10% puli syg. + 100% puli tła  →  S/B = 1:1.0
               Sygnał=21929   Tło=21929

────────────────────────────────────────────────────────────────────────

  MODEL: XGBoost
    n_estimators      = 3000  (best iter: 1996)
    learning_rate     = 0.05     max_depth         = 4
    min_child_weight  = 5        subsample         = 0.8
    colsample_bytree  = 0.8      gamma             = 1.0
    reg_alpha         = 0.1      reg_lambda        = 1.0
    scale_pos_weight  = 0.34

────────────────────────────────────────────────────────────────────────

  ROC AUC             : 0.9731
  Average Precision   : 0.9655

  Cut @ max-F1        : 0.230   (F1 = 0.923)
  Cut @ 95% bkg rej.  : 0.530   →  signal eff. = 0.840
  Cut @ 99% bkg rej.  : 0.886   →  signal eff. = 0.480

────────────────────────────────────────────────────────────────────────

  Classification report @ 95% bkg rejection threshold:

              precision    recall  f1-score   support

  Background       0.96      0.88      0.92     21929
        Muon       0.89      0.96      0.92     21929

    accuracy                           0.92     43858
   macro avg       0.92      0.92      0.92     43858
weighted avg       0.92      0.92      0.92     43858
```

Pełna numeryczna tabela ważności cech (XGBoost gain)
Wszystkie 36 cech posortowane malejąco | ρ(label) = korelacja Spearmana z etykietą (IsMuon)

Ranki 1 – 18

#	Feature	XGB Imp.	Cumul.	ρ(label)	Group
1	Radial_HCal_Fraction	0.30361	0.304	+0.348	HCal
2	Hcal_transverse	0.08809	0.392	+0.332	HCal
3	Ecal_Shape_radius	0.06666	0.458	-0.250	ECal
4	Hcal_Shape_theta_width	0.05839	0.517	-0.214	HCal
5	HCal_Number	0.05629	0.573	-0.401	HCal
6	HCal_Density	0.03639	0.609	+0.278	HCal
7	Hcal_angular_asym	0.03464	0.644	-0.303	HCal
8	Hcal_Shape_z_width	0.03321	0.677	-0.145	HCal
9	disp_ratio	0.02749	0.705	-0.151	Scalar/Combined
10	Hcal_Shape_dispersion	0.02605	0.731	+0.262	HCal
11	ECal_EoverP	0.02441	0.755	-0.218	ECal
12	HCal_Energy	0.02413	0.779	+0.075	HCal
13	Hcal_Shape_x_width	0.02277	0.802	+0.322	HCal
14	ECal_Energy	0.01967	0.822	-0.218	ECal
15	Total_CalEnergy	0.01956	0.841	-0.159	Scalar/Combined
16	ECal_HCal_ratio	0.01823	0.860	-0.184	ECal
17	radius_ratio	0.01315	0.873	-0.154	Scalar/Combined
18	Hcal_Shape_radius	0.01287	0.886	+0.185	HCal

Ranki 19 – 36

#	Feature	XGB Imp.	Cumul.	ρ(label)	Group
19	HCal_Fraction	0.01284	0.898	+0.184	HCal
20	ECal_Number	0.01121	0.910	-0.129	ECal
21	Hcal_Shape_y_width	0.01117	0.921	+0.052	HCal
22	HCal_EoverP	0.01047	0.931	+0.105	HCal
23	Hcal_long_trans_ratio	0.01029	0.942	-0.162	HCal
24	Ecal_Shape_dispersion	0.00821	0.950	-0.244	ECal
25	Hcal_Shape_phi_width	0.00765	0.957	+0.169	HCal
26	ECal_Density	0.00627	0.964	-0.194	ECal
27	Ecal_Shape_theta_width	0.00618	0.970	-0.235	ECal
28	Ecal_Shape_phi_width	0.00588	0.976	-0.119	ECal
29	Ecal_transverse	0.00500	0.981	-0.175	ECal
30	Ecal_angular_asym	0.00401	0.985	-0.063	ECal
31	Ecal_Shape_x_width	0.00356	0.988	-0.110	ECal
32	Ecal_Shape_y_width	0.00330	0.992	-0.191	ECal
33	transverse_ratio	0.00306	0.995	-0.050	Scalar/Combined
34	Ecal_Shape_z_width	0.00299	0.998	-0.150	ECal
35	Ecal_long_trans_ratio	0.00230	1.000	-0.146	ECal
36	z_width_ratio	0.00000	1.000	-0.054	Scalar/Combined

ECal features | HCal features | Scalar / Combined